

RODRIGO GOMES DE ARAÚJO

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SUMMARY

MSc Computer Science student focused on building **secure and reliable software** for the people and businesses who depend on it every day, from financial services to large-scale digital platforms. Enjoys **working across teams** and exchanging ideas openly, is quick to **adopt new approaches and tools**, values **clear communication** and **high-quality delivery** and thrives in **fast-paced learning environments**.

EDUCATION

Master's <i>Computer Science</i>	Sep 2025 – Expected Jul 2027 <i>University of Porto (FCUP)</i>
Bachelor's <i>Computer Science and Engineering</i>	Sep 2022 – Jul 2025 <i>University of Porto (FEUP)</i>

PROJECTS

Cloud-Native Hospital Backend | Python, Flask, Google Cloud, Artillery

- **Cloud-native REST API** on **Google Cloud** for patient records, admissions, clinical progress, media and patient-doctor questions.
- **Server-side access control**, with ownership rules enforced in the service layer.
- **Storage split is by access pattern**, structured records in **Firestore**, binary files as **blobs in Cloud Storage** with only their metadata indexed and precomputed aggregates in dedicated **cache collections**.
- Heavy aggregations moved to **scheduled serverless functions** with cached views, so derived endpoints respond as fast as plain reads.
- Validated with smoke tests and **Artillery load testing** (6,480 requests, **zero errors**, p95 under 185 ms).

PoW Blockchain for Decentralized Auctions | Rust, Tokio, LibP2P, Dalek, Docker

- Extensive **research on the state of the art of Bitcoin**, Ethereum and Solana and investigation regarding the security and resilience to failure of these systems.
- Developed a PoW Blockchain with a Kademlia base for **decentralized network construction**.
- **Client for auctions** using the blockchain layer as a public ledger.
- Docker **adversarial network simulation** with clients and bootstrap, well-behaved and malicious nodes.
- Developed for the Data Systems Security class and graded 19.1/20.

Self-Hosted Personal Portfolio | Next.js, React, TypeScript, Tailwind, Docker, Cloudflare, Tailscale

- **Personal portfolio** built with **Next.js and React in TypeScript**, styled with Tailwind and shadcn/ui.
- **Containerized with Docker** and published to Docker Hub, deployed and managed through **Portainer** on a personal home server.
- Exposed publicly via **Cloudflare Tunnel** and administered remotely over a private **Tailscale** network.

RESEARCH

CaST (Cryptographically-aware Session Types) <i>Master's Thesis</i>	Jul 2026 – Present <i>FreeST, Haskell, Type Systems, Applied Cryptography</i>
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- Extending the expressiveness of session types to capture **cryptographic resource usage**, so that its misuse in protocols is detected and rejected at compile-time.
- **Resource tracking composes across nested subprotocols** of arbitrary depth and is preserved under sequential composition and recursion.

AWARDS

2nd Place in IEEE RetroJam 2025 IEEE UP Student Branch	Oct 2025
• Built a full game in Rust from scratch in 48 hours, learning how to reduce a problem to its core requirements .	

TECHNICAL SKILLS

Programming Languages: Rust, Haskell, C/C++, OCaml, Java, SQL
Frameworks & Tools: Git, UNIX, Tokio, Docker, LibP2P, Dalek, Plotly, POSIX, STM, gRPC, PostgreSQL
Spoken Languages: Portuguese (Native), English (B2 First, Cambridge), Spanish (Proficient)